

Literature Review:

Paper Title	Method	Dataset	Advantages	Limitations	Research Gap
SWE-InfraBench: Evaluating Language Models on Cloud Infrastructure Code	Introduces SWE-InfraBench, a benchmark for evaluating LLMs on incremental Infrastructure-as-Code (IaC) editing tasks in AWS CDK repositories using natural-language instructions and automated unit tests.	100 instruction-based tasks collected from 34 real-world AWS CDK repositories.	Focuses on realistic enterprise workflows; supports imperative IaC; uses automated validation and real-world repositories.	Restricted mainly to AWS CDK. Does not fully analyze deployment reliability, security compliance, or production environments.	An integrated framework that supports multiple IaC formats and jointly evaluates deployment reliability, security validation, and real-world infrastructure behavior is still lacking.
Multi-IaC-Eval: Benchmarking Cloud Infrastructure as Code Across Multiple Formats	Proposes Multi-IaC-Bench, a benchmark for IaC generation and mutation across CloudFormation, Terraform, and CDK using natural-language requests.	Synthetic triplets consisting of an initial template, a modification request, and an updated template generated from public repositories.	Covers multiple IaC languages; supports both generation and mutation; introduces standardized evaluation metrics.	Relies heavily on synthetic data and semantic evaluation through LLM judges and static-analysis tools.	More realistic evaluation frameworks that validate infrastructure in production-like environments with deployment, security, and operational feedback are still required.
Deployability-Centric Infrastructure-as-Code Generation: Fail, Learn, Refine, and Succeed through LLM-Empowered	Introduces IaCGen, an iterative framework that refines generated templates using format verification, syntax checking, and	DPIaC-Eval benchmark containing 153 real-world scenarios across 58 AWS services.	Moves beyond syntax by emphasizing deployability; mirrors DevOps workflows; incorporates iterative feedback and	Requires repeated deployment attempts and human intervention, increasing cost and reducing scalability. User-intent	Scalable autonomous systems capable of improving deployment success, security compliance, and user-intent

DevOps Simulation	live deployment feedback.		human guidance.	alignment and security compliance remain weak.	satisfaction without extensive human involvement remain underexplored.
RIVA: Leveraging LLM Agents for Reliable Configuration Drift Detection	Proposes RIVA, a multi-agent architecture with verifier and tool-generation agents that collaborate through cross-validation and tool-history tracking.	Evaluated on the AIOpsLab benchmark containing infrastructure localization and drift-detection tasks.	Improves robustness against faulty tools and enhances configuration -drift detection accuracy.	Focuses mainly on configuration -drift detection and is evaluated only in benchmark settings.	Unified multi-agent frameworks that combine infrastructure generation, repair, deployment validation, and drift detection in real cloud environments are still needed.
TerraProbe: A Layered-Oracle Framework for Detecting Deceptive Fixes in LLM-Assisted Terraform Security Repair	Introduces TerraProbe, a five-layer oracle framework for evaluating LLM-generated Terraform security repairs.	288 first-pass repairs generated across 68 real-world Terraform modules and 28 injected-defect modules.	Evaluates planning validity and security intent in addition to scanner results; includes statistical analysis and multi-model comparisons.	Limited to Terraform security repair and emphasizes evaluation rather than automatic secure repair generation.	General-purpose frameworks capable of automatically generating and validating secure repairs across multiple IaC technologies and deployment platforms remain insufficient.